CSCI 4022 Final Project Writeup

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1 Abstract

Accurate mushroom classification is crucial for safe consumption, yet traditional methods based on physical characteristics pose challenges for novice foragers. This study investigates the potential of advanced data science techniques to improve edibility prediction. We leverage a rich dataset encompassing over 61,000 mushrooms across 173 species, described by 20 attributes including size, shape, texture, color, and habitat. K-Means clustering and Gaussian Mixture Models (GMMs) are employed to identify key parameters influencing edibility. K-Means clustering ideally done with k=2 is used for initial exploratory analysis, aiming to group mushrooms based on features and assess the feasibility of a binary classification (edible vs. poisonous) based on morphological characteristics. We then utilize GMMs to model the distribution of these features within the two predicted classes (edible and poisonous), allowing for a more nuanced understanding of the underlying data structure. This research contributes to the development of data-driven solutions for ecological classification tasks.

Figure 1: Gilled Mushroom

2 Introduction

Mushrooms represent a taxonomically diverse kingdom within the fungal domain, playing a vital ecological role in nutrient cycling and decomposition [1]. However, their consumption necessitates accurate identification due to the presence of potent toxins in many species [2]. Traditional classification methods rely on visual inspection of physical characteristics, often documented in field guides and taxonomic keys

[3]. However these methods present challenges for people who lack the requisite knowledge and experience to confidently distinguish edible from poisonous varieties. Additionally, look-alike species with highly similar morphological features can further complicate identification [4].

2.1 Motivation

Advancements in data science offer novel approaches for tackling complex classification tasks. We wanted to use data science algorithms for pattern recognition and uncovering hidden relationships within large datasets. This research is motivated by the potential of data science to revolutionize mushroom classification, offering a more objective and potentially more accessible solution compared to traditional methods. By leveraging advanced data science techniques, we aim to identify a minimal set of key features that can be used to accurately predict a mushroom's edibility. This not only empowers novice foragers but also paves the way for the development of automated identification tools that can be readily deployed in the field.

2.2 Data

This study utilizes a publicly available dataset from Kaggle, containing information on over 61,069 hypothetical mushrooms across 173 distinct species [5]. Each mushroom instance is characterized by 20 attributes, encompassing both categorical and numerical features. Categorical features include cap shape, cap surface, cap color, gill attachment, gill spacing, gill color, stem shape, stem root, stem surface, stem color, veil type, veil color, has-ring, ring type, spore-print color, and habitat. Numerical features include cap diameter, stem height, and stem width (all in centimeters). The comprehensive nature of this dataset, encompassing a wide variety of species and detailed morphological descriptions, provides a robust foundation for our analysis.

3 GMM

As discussed, the data had 20 fields consisting of cap-color, cap-shape, etc. many of which are categorical data. There were only 3 fields which were numerical (cap diameter, stem height and stem width). GMMs works by applying the normal distribution to data which means that the majority of the fields can't be used as it is text/categorical. Since we are missing out on a lot of data for this model, we knew GMM wouldn't be the best model. Despite this issue, we wanted to see how much the numerical data actually contributed to separating poisonous and edible mushrooms.

We first tested GMM with 2 clusters, trying the most basic poisonous vs edible separation. As you can see in the image below, the top plot is the GMM 2 clusters, while the bottom plot is the actual poisonous vs edible distribution. From this, we can see that 2 clusters was somewhat close to the actual distribution but still far from being a reliable way to predict if mushrooms are poisonous or not.

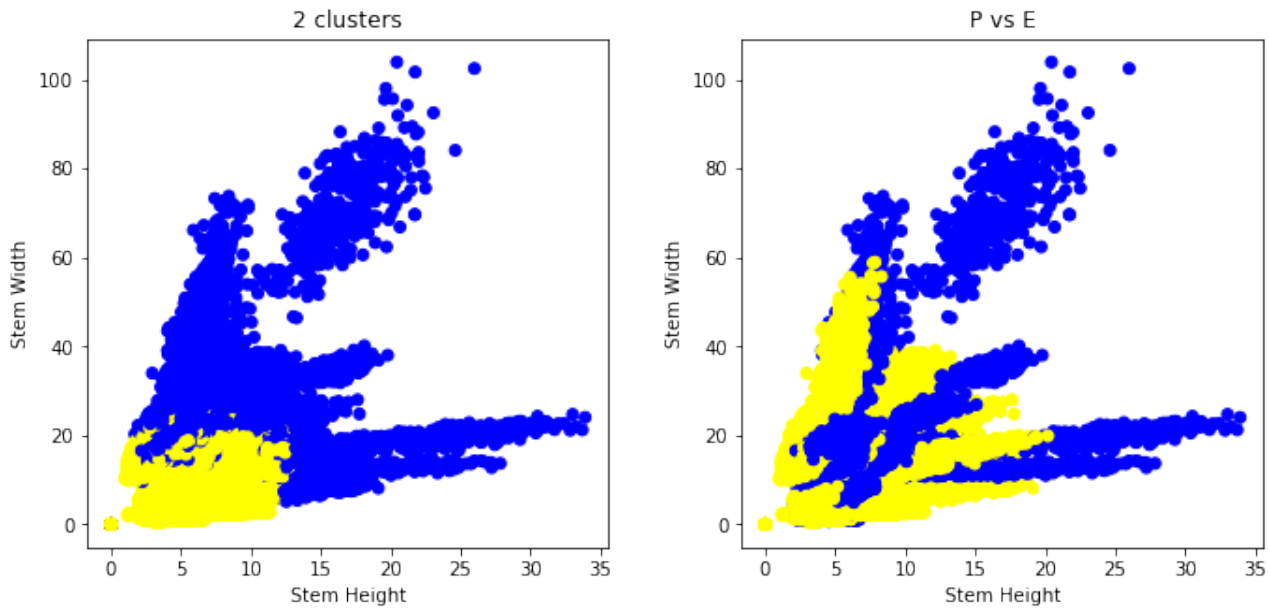

Figure 2: 2 Cluster and Actual Poisonous vs Edible based on Stem Height and Stem Width

We decided to check the data against more clusters thinking that there might be more patterns within the numerical data. And if we separated the current clusters enough, we thought we might eventually get clusters that are only poisonous or only edible. As you can see below, the first image shows the GMM model with 8 clusters, and the plot below shows the poisonous vs edible distribution again. And if we were to match the colors with the class, it would model the mushroom data better than the 2 clusters did.

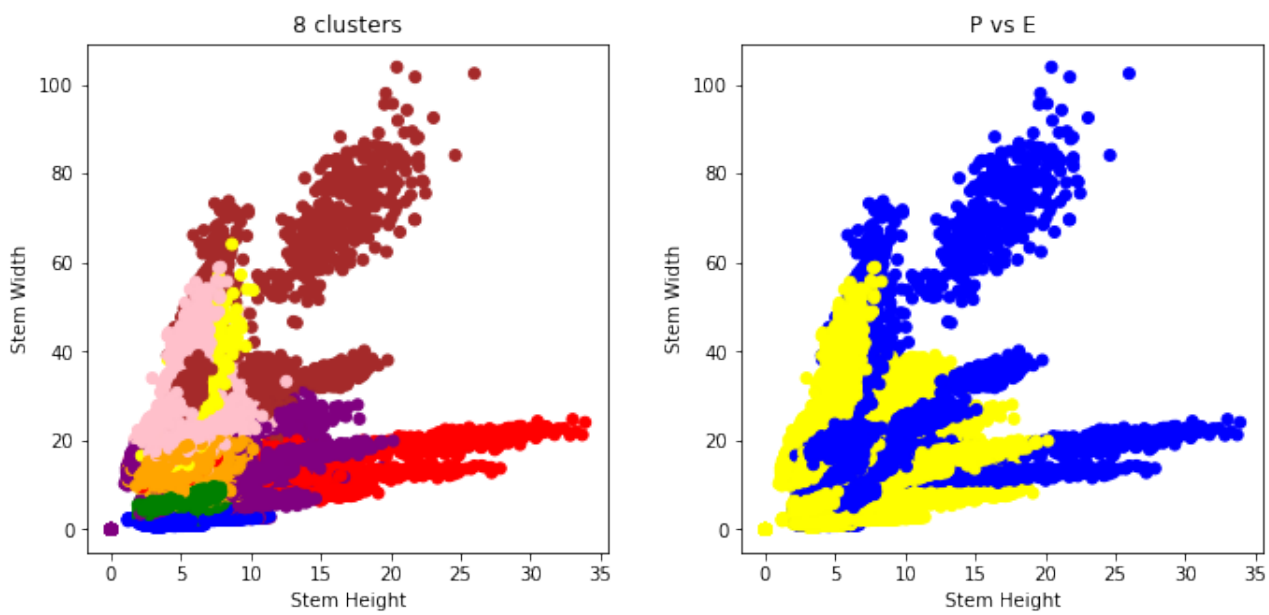

Figure 3: 8 Cluster and Actual Poisonous vs Edible based on Stem Height and Stem Width

At first, we thought our model was getting better with more clusters. However, through more testing we found that GMM in our case is not reliable enough to correctly predict the classification. We did this by finding the percentages of uniformity in the clusters (so basically how much of the cluster is poisonous or edible, whichever is greater). For example, if a cluster contained 70 percent poisonous mushrooms and 30 percent edible mushrooms, the uniformity value for the cluster would be 70 percent. For the 8 clusters, we got the values in the below image. If we found a new mushroom that would be placed in cluster 4, we would be able to tell whether it is poisonous or edible. However, if we found a mushroom in cluster 1, it would be a 50-50 chance to determine. The average of these values is about 65 percent, so that is not very good.

```
[0.519309262166405,
 0.638946352227279,
 0.7092103662246015,
 1.0,
 0.5619961078676675,
 0.600428494911623,
 0.5720355565494494,
 0.7262783006868482]
```

Figure 4: Probability of each of the 8 clusters being all poisonous or edible

We decided to run the above test against multiple clusters and see how the average plays out. As you can see in the image below, the percentage goes up from 2 to 5 clusters from 54 percent to 68 percent, however after those values there is a decline from 5 clusters to 11 clusters. From this, we can see that the highest average value we got was 68 percent for 5 clusters. Ideally, we would want this value to be 100 percent as we want all the clusters to be only one classification or the other so we would be able to confidently say whether a new mushroom is poisonous or not. However, since we were only able to get 68 percent at best, this shows that GMM with these numerical fields is not sufficient enough.

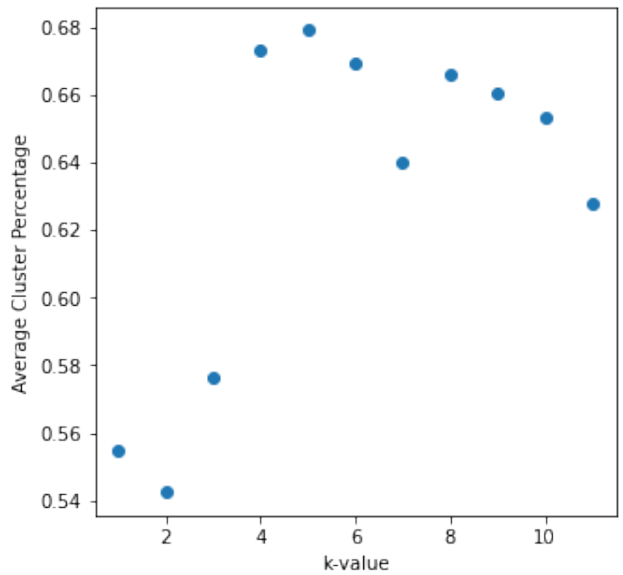

Figure 5: Finding Highest Average Cluster Percentage based on k-value

4 K-Means

K-means clustering is a clustering algorithm that attempts to assign n different data points or observations into k clusters. Each cluster has unique properties like a mean location and variance of all the points. K-means clustering is a computationally NP-hard problem but there exist heuristic algorithms that can converge to local optimal solutions.

The standard algorithm we use picks an initial set of k data points. These points are chosen at random. Then comes the assignment step. The assignment step assigns each observation/data point to a cluster with the least distance. The update step then recalculates the properties of the centroids such as the location/mean.

There were only three features that we could do K-means clustering on. These three features were numerical. If we wanted to implement K-means on other features such as categorical we would have had to one hot encode the data. Additionally, we'd have to come up with a distance metric so that each feature had a specific significance. We tried attempts at one hot encoding the data and and creating a distance measure between categorical data however we ran into errors with our python code. And for the sake of space, we decided to exclude that part of k-means from our writeup and code.

Below we clustered with 2 clusters. We will include another graph showing the actual poisonous vs edible clusters.

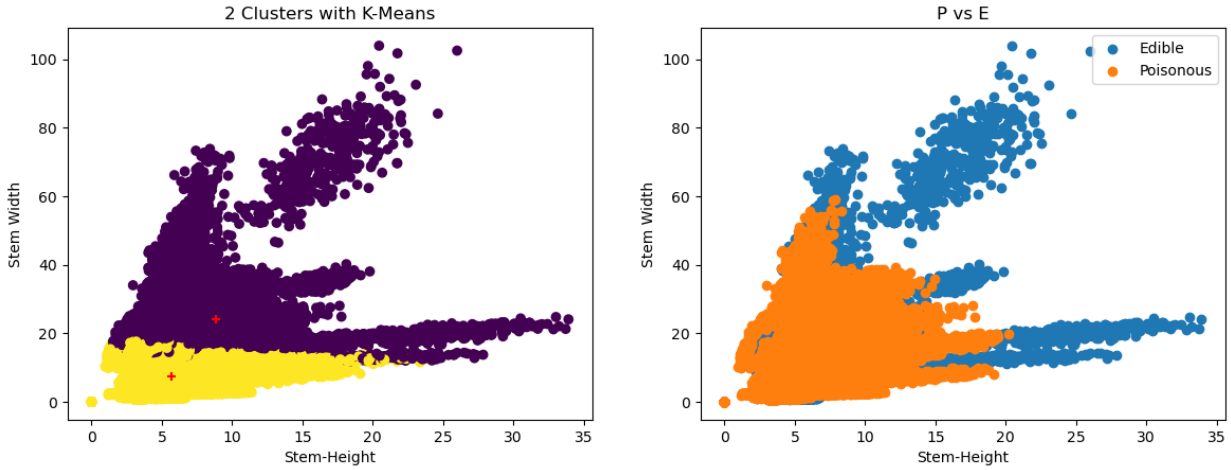

Figure 6: 2 cluster with K-means vs actual poisonous vs edible graph for stem-height vs stem width

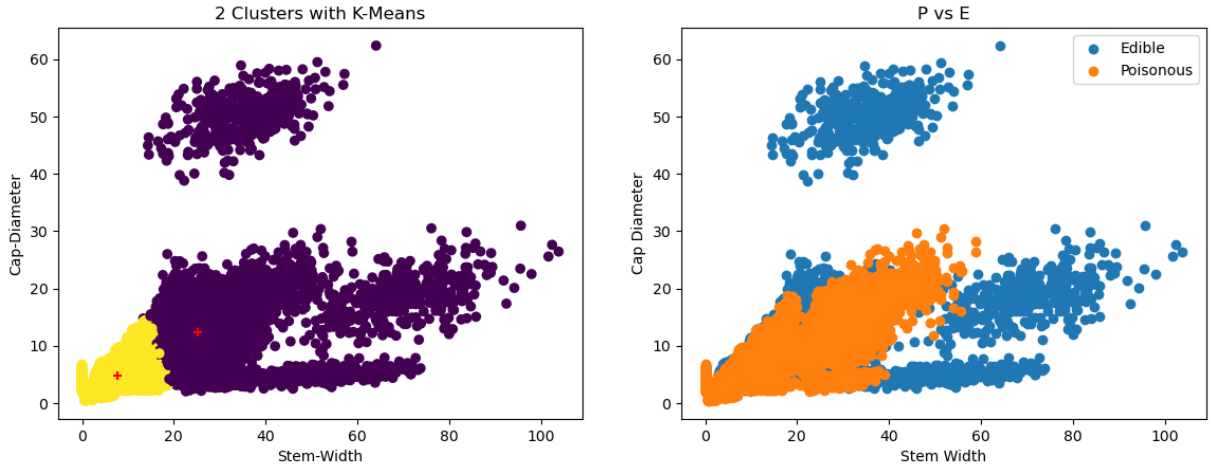

Figure 7: 2 cluster with K-means vs actual poisonous vs edible graph for stem-width vs cap diameter

We also clustered with 5 clusters. Again we will include another graph showing the actual poisonous vs edible clusters.

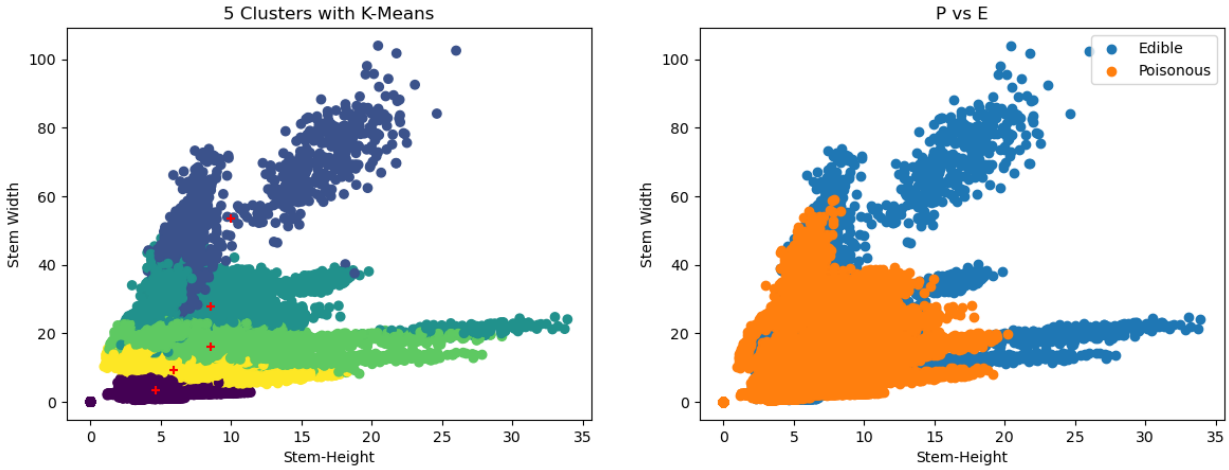

Figure 8: 5 cluster with K-means vs actual poisonous vs edible graph for stem-height vs stem width

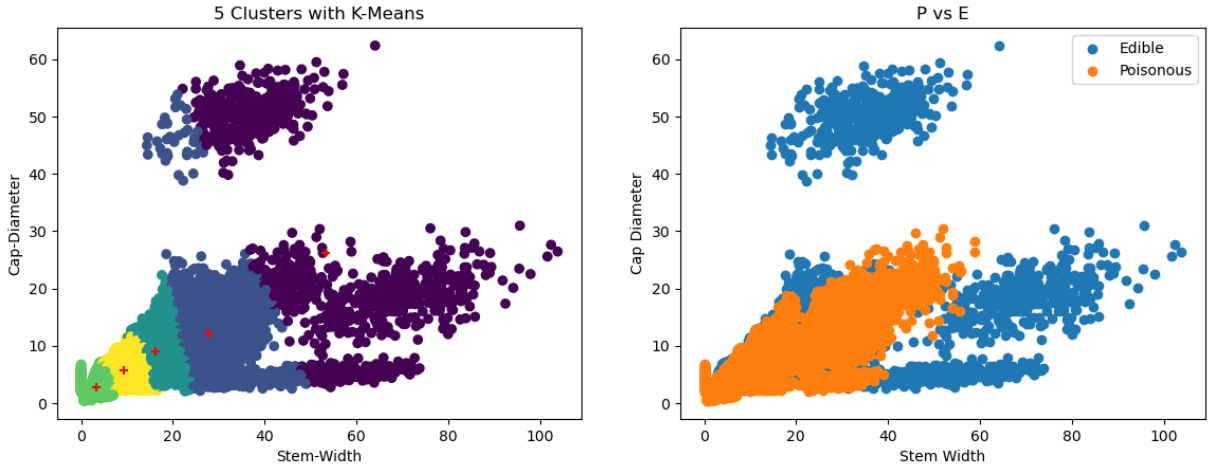

Figure 9: 5 cluster with K-means vs actual poisonous vs edible graph for stem-width vs cap diameter

We clustered with 5 clusters because when we created an error plot of the clusters, we found that the elbow of the error plot was at 5. Below is the error plot:

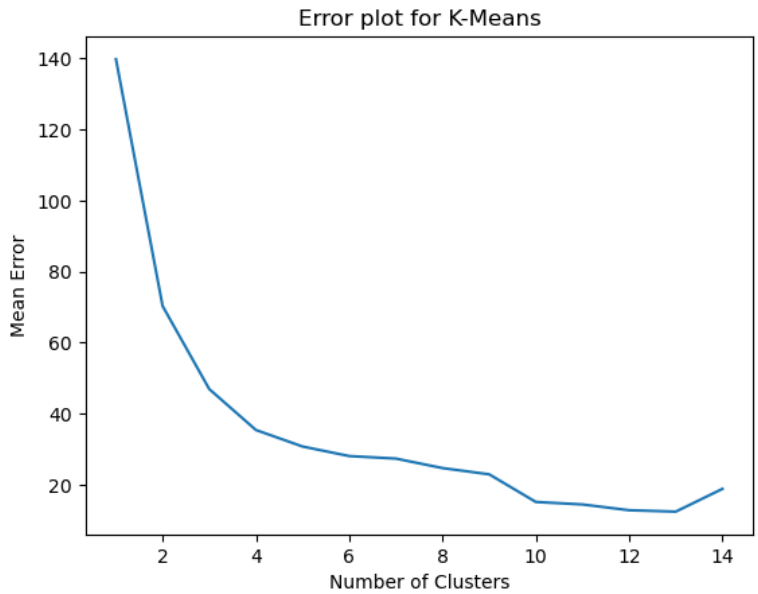

Figure 10: Mean error of all k-means clustering

Our analysis showed that clustering is not a good metric for predicting poisonous vs edible mushrooms. There seems to be an extra dimension that we aren't accounting for. Unfortunately, including the categorical data in our k-means clustering was unsuccessful so we aren't able to do an analysis on that. Fortunately we did find that all poisonous mushrooms are found with a stem width of less than 60 units, stem height of less than 21 units, and a cap diameter of less than 31 units. Any other mushroom that is out of these ranges we can guarantee are edible.

5 Conclusion

From our exploration of the K-Means clustering and Gaussian Mixture Models (GMMs) for mushroom classification, we revealed multiple limitations in accurately predicting mushroom edibility. This was further due to limiting our models to rely on solely numerical data. The predominately categorical parameters of our mushroom dataset (3 unique parameters) were insufficient to effectively separate mushrooms. We did manage to attain a 68% accuracy in classifying edibility, but a 32% chance of serious injury or death is not high enough for real world application. Due to technical challenges we were unable to incorporate categorical data in our study. While we are unable to confidently say anything about edibility of all mushrooms, we are able to explore the nuance of ecological classification. From this insight, we can highlight the importance of additional dimensions beyond the dataset. For example, factors such as biochemical composition, ecological interactions, and genetic characteristics may play hidden roles in mushroom edibility but were not addressed in our analysis. In conclusion, while data science techniques hold potential for enhancing mushroom classification, our study suggests that a multidisciplinary approach is essential for developing more accurate and reliable solutions.

References

[1] Moore, D. (2008) *Introduction to Fungi (4th ed.)*, Benjamin Cummings.

[2] Benjamin, D. R. (1995) *Mushrooms: poisons and hallucinogens*, Dover Publications.

[3] Kuo, M. (2000) *Mushrooms of the Midwest [Field Guide]*, University of Illinois Press.

[4] Beug, M. W., Besl, H., & Obermayer, W. (2017) *Mushrooms of Central and Continental Europe (B. Oertel & D. Arora, Eds.)*, Princeton University Press.

[5] Naik, Vishal P. *Mushroom Classification: Edible or Poisonous?*, Kaggle, 8 Mar. 2024, www.kaggle.com/datasets/vishalpnaik/mushroom-classification-edible-or-poisonous.